

This document provides instructions on freeware installation on your own Notebook computer or MacBook and running a simple demo.
Colab users will not need to install the modules.

(A) Instructions for Installing Python, Pip, Tensorflow, jupyter, Keras, matplotlib, sklearn, PyTorch, OpenCV, Colab

1. Python

There are two versions of Python, python2 and python3. Now, most of developers are using Python3 rather than python2. Hence, you only need to install **Python 3** on your PC.

1.1 Download

The latest tensor-supported Python version is 3.7. You are recommended to install Python 3.7 (e.g. python 3.7.10) which can support Tensorflow well.

You can download it here:

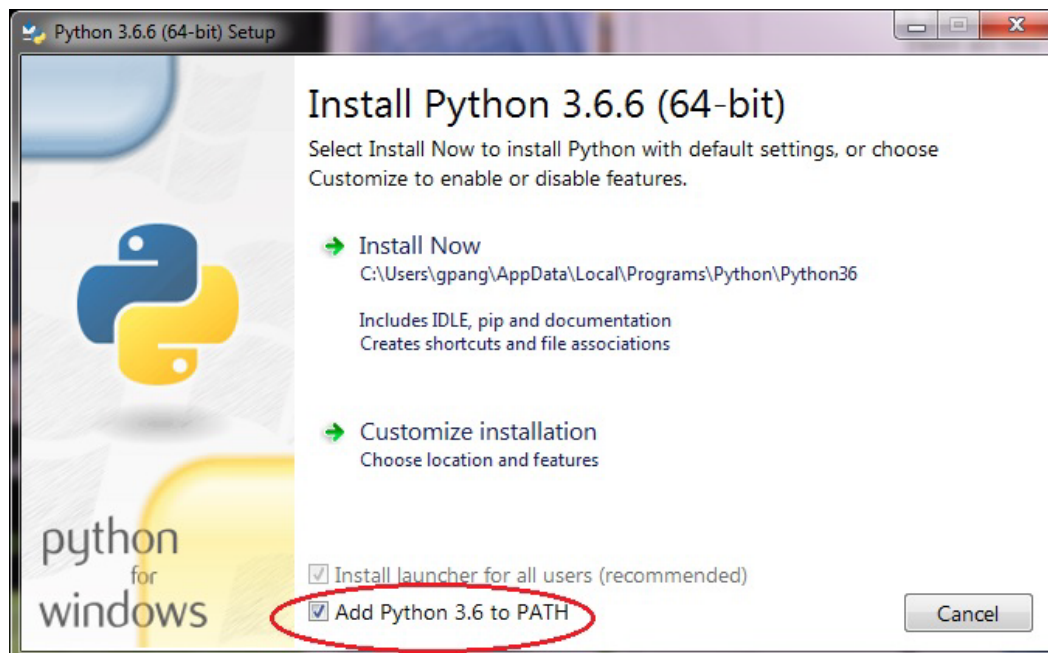
<https://www.python.org/downloads/>

Just go to this page to select. For Windows, download the "[Windows x86-64 executable installer](#)" to your PC. For MacOSX, choose the appropriate installer.

1.2 Installation (the example is based on python 3.6.6)

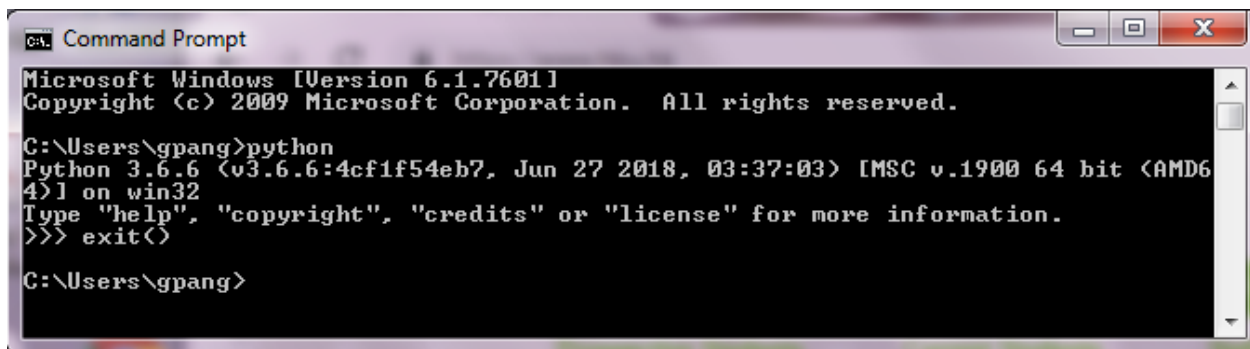
After downloading the installer, you can just double click the file and install Python3 following the installation steps.

It is important to click the button "Add Python 3.6 to Path" before you click "Install Now".





1.3 To verify the installation, go to your **Command Prompt** and type **python**. If you want to exit the python command, you can type in "exit()" and you can quit the python program.



1.4 Online tutorial of python: <https://docs.python.org/3.6/tutorial/index.html>

In Mac, may require "alias python=/usr/local/bin/python3" to setup properly.

Sublime Text is a good editor for .py file.

2. pip

pip is a package management system in python, which can be used to install and manage software packages (e.g. numpy, pandas and tensorflow). Hence, you must install pip when using python.

2.1 Installation: <https://pip.pypa.io/en/stable/installation/>

a) Get file called "get-pip.py".

bootstrap.pypa.io/pip/3.6/get-pip.py to get the file.

b) If you already have the file, in the Command Prompt, change directory to go to the directory (e.g. Install_Python) which contains the file (get-pip.py).

e.g. cd \Install_Python

- c) Type in "**python get-pip.py**" in the command window.
- b) The program should be running and will install well.

```

C:\Windows\system32\cmd.exe
C:\Users\gpang\AppData\Local\Programs\Python\Python36>python get-pip.py
Collecting pip
  Downloading https://files.pythonhosted.org/packages/5f/25/e52d3f31441505a5f3af
41213346e5b6c221c9e086a166f3703d2ddaf940/pip-18.0-py2.py3-none-any.whl (1.3MB)
    100% |#####| 1.3MB 4.9MB/s
Collecting wheel
  Downloading https://files.pythonhosted.org/packages/81/30/e935244ca6165187ae8b
e876b6316ae201b71485538ffac1d718843025a9/wheel-0.31.1-py2.py3-none-any.whl (41kB)
    100% |#####| 51kB 3.9MB/s
Installing collected packages: pip, wheel
  Found existing installation: pip 10.0.1
  Uninstalling pip-10.0.1:
    Successfully uninstalled pip-10.0.1
  The script wheel.exe is installed in 'C:\Users\gpang\AppData\Local\Programs\Py
thon\Python36\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warn
ing, use --no-warn-script-location.
Successfully installed pip-18.0 wheel-0.31.1
C:\Users\gpang\AppData\Local\Programs\Python\Python36>

```

2.2 Check pip

- a) Type in "**pip --version**" from Command Prompt to show the version of pip
- b) Type in "**pip install pandas**" to check whether the package pandas can be installed well.

```

Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\gpang>pip --version
pip 18.0 from c:\users\gpang\appdata\local\programs\python\python36\lib\site-pac
kages\pip (python 3.6)

C:\Users\gpang>pip install pandas
Collecting pandas
  Downloading https://files.pythonhosted.org/packages/0e/67/def5bfa4d3324fdb890
48889ec523c0903c5efab1a64c8dbe0ac8eec13c/pandas-0.23.4-cp36-cp36m-win_amd64.whl
(7.7MB)
    100% |#####| 7.7MB 2.1MB/s
Collecting pytz>=2011k (from pandas)
  Downloading https://files.pythonhosted.org/packages/30/4e/27c34b62430286c6d591
77a0842ed90dc789ce5d1ed740887653b898779a/pytz-2018.5-py2.py3-none-any.whl (510kB)
    100% |#####| 512kB 5.7MB/s
Collecting numpy>=1.9.0 (from pandas)
  Downloading https://files.pythonhosted.org/packages/fb/7d/f8b97d97809f184d90fa
f320fa8e2e7eac994844c5e6c57adbed1283e9e9/numpy-1.15.1-cp36-none-win_amd64.whl (1
3.5MB)
    100% |#####| 13.5MB 1.2MB/s
Collecting python-dateutil>=2.5.0 (from pandas)
  Downloading https://files.pythonhosted.org/packages/cf/f5/af2b09c957ace60dcfac
112b669c45c8c97e32f94aa8b56da4c6d1682825/python_dateutil-2.7.3-py2.py3-none-any.
whl (211kB)
    100% |#####| 215kB 6.4MB/s
Collecting six>=1.5 (from python-dateutil>=2.5.0->pandas)
  Downloading https://files.pythonhosted.org/packages/67/4b/141a581104b1f6397bfa
78ac9d43d8ad29a7ca43ea90a2d863fe3056e86a/six-1.11.0-py2.py3-none-any.whl
Installing collected packages: pytz, numpy, six, python-dateutil, pandas
Successfully installed numpy-1.15.1 pandas-0.23.4 python-dateutil-2.7.3 pytz-201
8.5 six-1.11.0

C:\Users\gpang>

```

2.3 To upgrade pip (to version 18.1)

Type in “**python –m pip install -- upgrade pip**” in the Command Window.

3. tensorflow

You can use pip to install tensorflow. Some information can be found in https://www.tensorflow.org/install/install_windows

3.1 Installation

Type in "**pip install tensorflow==1.5.0**" in command window.

3.2 Check tensorflow

- a) type in 'python' in command window
- b) type in "import tensorflow as tf" in python window
- c) To show the tensorflow version

```
>>> import tensorflow as tf
>>> print(tf.__version__)
1.5.0
```

3.3 To validate the tensorflow:

```
>>> import tensorflow as tf
>>> hello = tf.constant('Hello, TensorFlow!')
>>> sess = tf.Session()
>>> print(sess.run(hello))
```

Other relevant commands that may be useful:

"pip uninstall tensorflow" in command window.

"pip install --upgrade tensorflow" in command window.

"exit()" to quit or exit.

Note that some commands in python requires higher version of tensorflow. Otherwise, error messages below will appear:

tensorflow-io 0.17.0 requires tensorflow<2.5.0,>=2.4.0, but you have tensorflow 1.5.0 which is incompatible.

The following will update the tensorflow version to 2.4.1. (My recommended version)

D:\> pip install tensorflow -io

If using version 2.4.1 :

```
>>> import tensorflow as tf
2021-04-15 19:42:35.462970: W tensorflow/stream_executor/platform/default/dso_loader.cc:60] Could not
load dynamic library 'cudart64_110.dll'; dLError: cudart64_110.dll not found
2021-04-15 19:42:35.463139: I tensorflow/stream_executor/cuda/cudart_stub.cc:29] Ignore above cudart
dLError if you do not have a GPU set up on your machine.
>>> print(tf.__version__)
2.4.1
```

To switch back to tensorflow 1.5.0, goto Command Window:

D:\> pip install tensorflow==1.5.0

4. jupyter (a very useful python editor)

Jupyter Notebook is a web-based interactive computational environment which can help to edit and debug python documents. *It is a very useful facility.* You can use pip to install jupyter.

4.1 Installation

From command window, type in "**pip install jupyter**".

4.2 To invoke:

From command window, type in "**jupyter notebook**".

For Mac Notebook, you may need to run the following in the Command Prompt in order to start the jupyter notebook properly:

> **pip install markupsafe==2.0.1**

5. Keras

From command window, type in "**pip install keras**".

6. matplotlib (plotting library)

From command window, type in "**pip install matplotlib**".

7. sklearn

From the Windows Command Prompt, install sklearn framework:

> **pip install numpy scipy scikit-learn**

8(a). PyTorch

Notes to install PyTorch using Pip, with Python 3.6 previously installed.

From the Windows Command Prompt:

> **pip install https://download.pytorch.org/whl/cu90/torch-1.0.0-cp36-cp36m-win_amd64.whl**

> **pip install torchvision**

See <https://pytorch.org/get-started/locally/> for more details.

8(b). OpenCV

From the Windows Command Prompt, type in "**pip install opencv-python**".

9. Colab (Free Google online GPU service)

You need a gmail account.

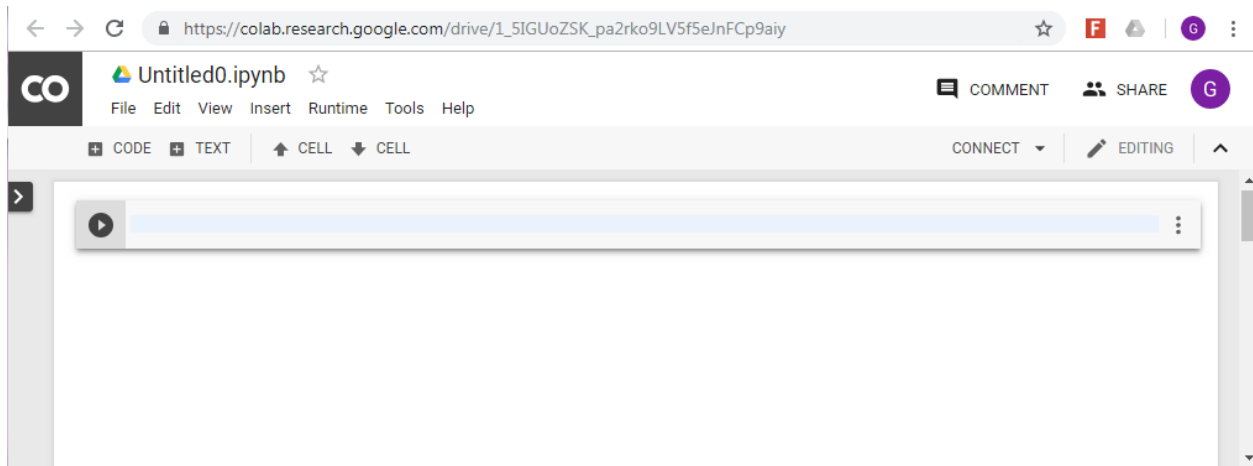
Go to <https://colab.research.google.com/notebooks/welcome.ipynb>

File → New notebook (Google Sign in)

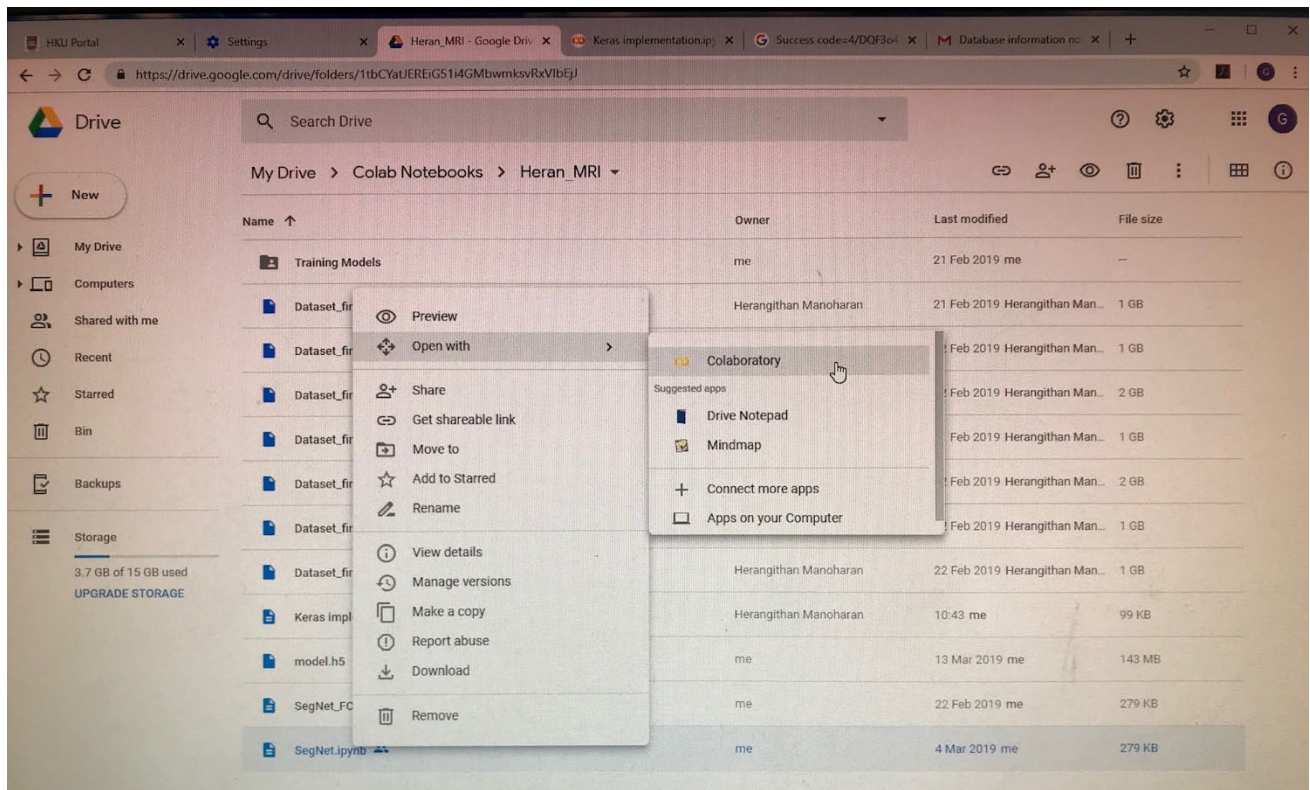
Runtime → Change runtime type (Go to Hardware Accelerator to choose “GPU”)

Click “CONNECT” (Upper Right Button)

You can then place your python code to your .ipynb environment to click the icon (**Run**) to execute.



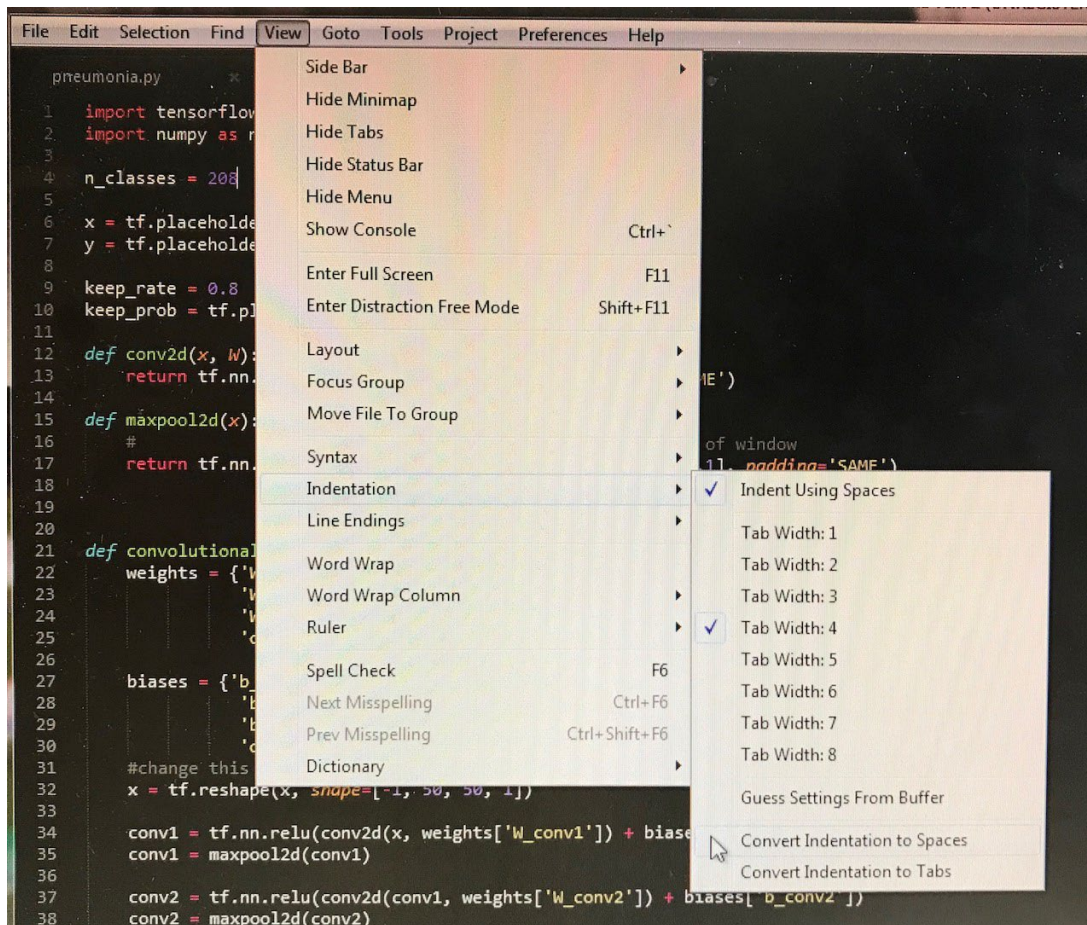
Suppose you have a .ipynb in your Google drive, and you want to open it. The following will take you to Colab with the opened project:



Very often, you need to mount your google drive to Colab.
See demo later.

Important Note:

- When you use text editor such as **Sublime** to write your python code.
Goto **View** → **Indentation** → **Indent Using Spaces**
Goto **View** → **Indentation** → **Convert Indentation to Spaces**



- To debug, you copy (**Ctrl-C to copy**) and paste the code to run in the command window.
- Yet, **in the command window, do not use Ctrl-V to paste**. You have to right click and select 'Paste'.
- Very often, you would copy your project directory path from the Windows, and then go to the command window to type '**cd**' and then **right click to select 'Paste'** to put in the directory path.
- The command window does not like a blank line feed, and hence it is best to debug using jupyter. You first read in your code from a file ****.py** and you can save it back. Then, you can run the entire file in the command window by "**python **.py**".

- Anyway, you are recommended to use either "**jupyter notebook**" or the **Colab** environment for the python programming. Homework submission is also either one of them (note that file extension is ***.ipynb**). In jupyter notebook or Colab, many modules are built-in and you do not have to install them one by one, like in the Command Prompt.

e.g.

```
import matplotlib.pyplot as plt
import numpy as np
import tensorflow as tf
import wave
```

10. Problem with running tensorflow in the Mac environment.

I find a lot of information about the problem of using TensorFlow in MacBook. It seems TensorFlow was built for Intel chip and there is a special version of tensorflow for Mac !

Search “Problem using TensorFlow in MacBook” in google and you can find a lot of warning.

Next, read below :

<https://medium.com/@tehreemansari/fixing-tensorflow-crashes-on-macbook-during-ml-model-training-40a1c5a9562f>

Hence, if you are a MacBook user, I recommend you to run my sample code (in .ipynb files) in Colab, rather than the jupyter notebook.